

Extended Notes toward Cosmological Dynamics under OMM–SBT–LIM

From Latent Structural Memory to Emergent Cosmological Organization

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Abstract

These extended notes explore possible cosmological consequences of the Oscillatory Mantle Model – Substantial Beating Theory – Latent Identity Model framework (OMM–SBT–LIM). The objective is not to claim a validated cosmological theory, but to organize a disciplined bridge between the substrate dynamics introduced in OMM–SBT and the latent memory/arbitration mechanisms explored in LIM.

We investigate whether persistent observable organization, non-terminal thermodynamic behavior, black-hole interface dynamics, effective gravitational amplification, and a variable cosmological term may emerge as compatible consequences of the same underlying structural framework.

Throughout the document, demonstrated computational results are explicitly separated from compatible theoretical extensions and from more speculative cosmological extrapolations.

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1 Introduction

OMM–SBT was originally introduced as a framework in which observable physical organization emerges from a deeper oscillatory substrate. Within this picture, observable dynamics are not necessarily treated as fundamental closed entities, but rather as stabilized manifestations of a deeper structural regime.

LIM later demonstrated, under strict observable aliasing and anti-cheat constraints, that observable similarity alone may fail to reconstruct identity, while coherent latent memory can restore reconstruction consistency.

The present document explores whether the same distinction between observable states, latent structure, and arbitration mechanisms may also illuminate broader cosmological questions.

The guiding question is therefore:

If the observable world corresponds to an emergent stabilized regime rather than to a fundamentally closed system, what cosmological dynamics become compatible with OMM–SBT–LIM?

Not claimed. These notes do not claim a validated cosmological model, a completed theory of black holes, or a finalized theory of quantum gravity. Their purpose is instead to organize a coherent extension of the OMM–SBT–LIM framework, identify structurally compatible cosmological consequences, and clarify which future calculations would be required for possible validation.

2 Inherited Foundations from OMM–SBT

Within OMM–SBT, the observable substrate is represented through the coupled state

$$\Psi_t = (X_t, \Phi_t, K_t), \quad (1)$$

where X_t denotes a mantle/amplitude field, Φ_t a phase or orientation field, and K_t a kinetic-memory field encoding persistent dynamical organization.

A minimal effective density proxy may be expressed as

$$\rho_{\text{eff}} = X^2 + \alpha_K K, \quad (2)$$

where the kinetic contribution allows memory-dependent structural effects to contribute to observable organization.

A current-like quantity is introduced as

$$J = |X| \nabla \Phi, \quad (3)$$

with associated vorticity-like structure

$$W = \nabla \times J. \quad (4)$$

Observable geometry is then treated as an effective consequence of substrate organization rather than as a strictly primitive entity. A schematic geometric proxy may therefore be written as

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + b|\nabla^2 X| + c|W| + d|J| + eK. \quad (5)$$

The precise functional form is not assumed to be fundamental here. Instead, the expression serves as a structural illustration of how effective geometry may emerge from density, curvature, flow coherence, vorticity, and kinetic persistence within the substrate dynamics.

This distinction is central for the cosmological extensions considered in the present work: observable spacetime geometry is treated as a stabilized effective regime generated by deeper organizational processes.

3 LIM Contribution: Observable Insufficiency and Latent Structure

LIM introduces a reconstruction-theoretic layer into the OMM–SBT framework.

Under strict observable aliasing, multiple candidate identities may share the same observable state:

$$\mathcal{O}_i = \mathcal{O}_j, \quad i \neq j. \quad (6)$$

Within the strict double-blind benchmark introduced in LIM, observable-only reconstruction collapses to chance-level recovery, whereas coherent latent memory restores reconstruction consistency.

Demonstrated. Within the tested formal regime, latent structure becomes functionally necessary for reliable reconstruction under true observable aliasing.

The broader implication is that observable information alone may be insufficient to fully characterize persistent organization.

This motivates the cosmological extension considered in the present work: the observable universe may itself fail to be dynamically self-sufficient when treated in isolation from deeper structural degrees of freedom.

4 Thermodynamic Persistence and Non-Terminal Observable Dynamics

A strictly closed observable thermodynamics generally tends toward terminal dissipation and equilibrium decay. OMM–SBT suggests a different possibility: observable organization may persist through continuous interaction with deeper substrate structure.

Let G_t denote an observable gradient or organization measure. A minimal persistence relation may be written as

$$\frac{dG_t}{dt} = -\gamma G_t + \eta F_{\mathcal{L}}(t), \quad (7)$$

where γ represents observable dissipation and $F_{\mathcal{L}}(t)$ denotes a latent feedback contribution originating from deeper structural dynamics.

If the latent contribution remains partially coherent rather than purely stochastic, observable gradients may avoid asymptotic collapse:

$$\liminf_{t \rightarrow \infty} G_t > 0. \quad (8)$$

Compatible hypothesis. Within the OMM–SBT–LIM perspective, this provides a possible route toward non-terminal observable dynamics without requiring the observable sector itself to constitute a fully closed thermodynamic system.

This interpretation does not eliminate entropy production locally. Instead, it suggests that persistent observable organization may be continuously sustained through deeper structural exchange processes.

5 Latent Sectors and Velocity Hierarchies

A recurring interpretation emerging from OMM–SBT–LIM is that the observable world may correspond to a slowed or stabilized regime of a deeper dynamical substrate.

Let v_o denote an effective observable propagation scale and v_ℓ a latent or pre-observable propagation scale:

$$v_\ell > v_o. \quad (9)$$

This inequality should not necessarily be interpreted as ordinary superluminal propagation within observable spacetime. Instead, the latent sector may possess a higher structural update rate relative to the emergent observable time coordinate.

This motivates an effective hierarchy of relaxation scales:

$$\tau_\ell < \tau_o, \quad (10)$$

where τ_ℓ denotes a latent relaxation timescale and τ_o an observable relaxation timescale.

Within this perspective, observable temporal dynamics may correspond to a partially coarse-grained projection of deeper substrate evolution.

Compatible hypothesis. The proto-calibration of latent memory obtained in LIM supports the possibility that part of the latent layer may be globally characterizable while remaining irreducible to a simple observable vector basis.

6 The Arbiter and the Limit of Geometrization

The LIM calibration scan suggested that latent memory may be partially characterizable, while the arbitration component resists direct reduction to a simple observable-vector calibration.

This motivates a cautious hypothesis:

The arbiter may correspond to a global coherence constraint rather than to an ordinary localized object embedded within observable spacetime.

Importantly, this interpretation should not be understood as introducing a supernatural entity or an external metaphysical mechanism. Within the present formalism, the term “arbiter” denotes only a structural selection principle governing compatible latent-observable consistency relations.

Not claimed. The present framework does not prove that the arbiter exists “outside spacetime” in any ontological sense. It indicates only that, within the current calibration regime, arbitration is not reducible to a directly observable geometric parametrization.

A possible hierarchy is therefore:

$$\text{observable geometry} \subset \text{latent structural memory} \subset \text{global arbitration constraint}. \quad (11)$$

In this interpretation, observable geometry may emerge from deeper structural consistency conditions rather than constituting the ultimate primitive layer of the theory.

7 Black Holes as Candidate Inter-Regime Attractors

Black holes may be interpreted, within this speculative extension, as candidate inter-regime attractors.

The guiding hypothesis is not that black holes constitute literal “portals” or exotic metaphysical objects, but rather that their extreme causal, geometric, and thermodynamic properties may identify transition zones between observable stabilization regimes and deeper substrate dynamics.

Speculative extension. Within this interpretation, a black hole may correspond to a region where observable geometry approaches maximal compression while latent kinetic organization remains only partially expressible through ordinary observable dynamics.

This motivates the schematic limiting condition

$$g_{\text{eff}} \rightarrow g_{\text{eff}}^{\text{crit}}, \quad v_{\text{eff}} \rightarrow v_\ell, \quad (12)$$

where v_{eff} denotes an effective local substrate update scale and v_ℓ a latent structural update regime.

The key idea is therefore not the destruction of structure, but a possible transition between dynamical regimes characterized by different effective timescales and geometric accessibility.

Possible interpretations include:

- black holes as latent-kinetic attractors;
- black holes as regulators of observable gradients;
- black holes as relaxation interfaces;
- black holes as remnants of high-speed substrate regimes;
- black holes as partial sinks of observable thermodynamic dissipation.

Under this interpretation, black holes would not necessarily terminate organization, but could instead participate in the redistribution or restructuring of observable and latent dynamical coherence.

Not claimed. These interpretations remain speculative and require independent mathematical, relativistic, and astrophysical validation.

8 Emergent Gravitation from Structural Density

If geometry is emergent rather than strictly primitive, then gravitation may depend on structural organization rather than observable mass alone.

Within the OMM-SBT framework, an effective gravitational intensity proxy may therefore be written schematically as

$$\mathcal{G}_{\text{eff}} = \lambda_\rho \rho_{\text{eff}} + \lambda_J |J|^2 + \lambda_W |W|^2 + \lambda_K K + \lambda_v v_{\text{eff}}^2, \quad (13)$$

where ρ_{eff} denotes effective structural density, J a current-like term, W a vorticity-like component, K a kinetic-memory contribution, and v_{eff} an effective substrate update scale.

This expression is not proposed as a final gravitational law. It is a minimal exploratory scaffold intended to test whether faster or more coherent substrate regimes can generate stronger effective curvature-like behavior.

The associated computational probe explored whether increasing v_{eff} produces measurable amplification of effective gravitational proxies.

Demonstrated. Within the tested numerical regime, increasing v_{eff} produced strong monotonic amplification of the effective gravitational proxy:

$$\text{corr}(v_{\text{eff}}, \mathcal{G}_{\text{eff}}) \simeq 0.978. \quad (14)$$

The simulated high-velocity regime produced an amplification factor of approximately

$$\mathcal{A}_{\text{grav}} \simeq 7.04 \quad (15)$$

relative to the lowest tested regime.

Compatible hypothesis. These results remain purely exploratory, but they support the compatibility of the following interpretation within OMM-SBT-LIM: a “faster” substrate regime may appear gravitationally stronger to the observable sector because increased structural update density generates stronger effective geometric organization.

9 Cosmological Constant as Observable-Scale Variable

In standard cosmology, the cosmological constant Λ is typically treated as a constant vacuum contribution to large-scale dynamics. Within OMM–SBT–LIM, it may instead be more natural to interpret the cosmological term as an emergent observable-scale variable.

A schematic effective relation may be written as

$$\Lambda_{\text{eff}}(t) = \Lambda_0 + \beta_1 \langle \rho_{\text{eff}} \rangle_t + \beta_2 \langle K \rangle_t + \beta_3 R_{\mathcal{O}}(t)^{-2} + \beta_4 C_{\mathcal{L} \rightarrow \mathcal{O}}(t), \quad (16)$$

where:

- $\langle \rho_{\text{eff}} \rangle_t$ denotes an averaged effective structural density;
- $\langle K \rangle_t$ an averaged kinetic-memory contribution;
- $R_{\mathcal{O}}(t)$ an effective observable horizon or accessible scale;
- $C_{\mathcal{L} \rightarrow \mathcal{O}}(t)$ a latent-to-observable coupling term.

The key idea is that the observable cosmological sector may inherit part of its large-scale dynamics from finite observable accessibility and from substrate organization rather than from a strictly fundamental constant vacuum term.

In this interpretation, the observable cosmological term becomes:

- emergent rather than strictly primitive;
- scale-dependent rather than strictly absolute;
- dynamically coupled to substrate organization.

Compatible hypothesis. This framework therefore supports the compatibility of a variable effective cosmological term without requiring the abandonment of large-scale relativistic cosmology within the observable regime.

10 Two Computational Probes

Two initial computational probes were implemented in order to test whether key OMM–SBT–LIM cosmological hypotheses remain internally compatible within simplified dynamical regimes.

These probes do not constitute empirical cosmological validation. They evaluate whether the proposed structural mechanisms produce the expected qualitative behaviors under controlled computational conditions.

10.1 Probe A: Persistence Dynamics

This probe evaluates whether observable gradients decay toward terminal dissipation or remain persistent when coupled to latent feedback.

The target observable is defined as

$$G_t = \text{Var}(X_t) + \text{Var}(K_t). \quad (17)$$

The simulated dynamics follow the schematic form

$$\frac{dG_t}{dt} = -\gamma G_t + \eta F_{\mathcal{L}}(t), \quad (18)$$

where γ denotes observable dissipation and $F_{\mathcal{L}}$ latent feedback.

A positive result would not demonstrate cosmological non-death, but would show that OMM-SBT-style latent feedback can sustain persistent organization in a simplified regime.

Demonstrated. Within the tested numerical conditions, observable gradients decay strongly in the no-feedback baseline:

$$G_{\text{ret}} \simeq 0.0154. \quad (19)$$

When structured latent feedback is introduced, gradients can instead remain persistent or become amplified. The strongest tested regime reaches

$$G_{\text{ret}} \simeq 1.0499, \quad (20)$$

corresponding to a persistence gain of approximately

$$\mathcal{A}_{\text{persist}} \simeq 68.04 \quad (21)$$

relative to the no-feedback baseline.

Compatible hypothesis. These results support the compatibility of OMM-SBT-LIM with non-terminal observable dynamics under latent feedback. They do not prove cosmological persistence, but they demonstrate that the proposed architecture can sustain persistent structure within a minimal computational model.

10.2 Probe B: Structural Gravitational Amplification

This probe evaluates whether effective gravitational intensity increases with structural density and latent velocity hierarchy.

The tested proxy is

$$\mathcal{G}_{\text{eff}} = f(\rho_{\text{eff}}, J, W, K, v_{\text{eff}}). \quad (22)$$

A positive result would support the compatibility of stronger effective gravitation with faster or denser substrate regimes.

Demonstrated. Within the tested numerical regime, the effective gravity proxy exhibits a strong monotonic relation with the latent velocity hierarchy:

$$\text{corr}(v_{\text{eff}}, \mathcal{G}_{\text{eff}}) \simeq 0.9781. \quad (23)$$

The highest tested velocity regime produces an amplification factor of approximately

$$\mathcal{A}_{\text{grav}} \simeq 7.04 \quad (24)$$

relative to the lowest tested regime.

Compatible hypothesis. These results support the internal coherence of the hypothesis that faster or denser substrate regimes may appear as stronger effective gravitational attractors to the observable sector.

Not claimed. Since the velocity hierarchy is explicitly included in the tested proxy, this result does not constitute an empirical discovery of a new gravitational law. It validates the behavior of the proposed structural proxy within the tested computational framework and does not replace general relativity.

11 Open Mathematical Problems

The present extension remains highly incomplete and should be understood as a structured exploratory framework rather than a finalized physical theory.

Several major mathematical and physical problems remain open:

- deriving a fully invariant metric formulation from the OMM–SBT substrate variables;
- establishing explicit relations between g_{eff} and standard relativistic curvature tensors;
- deriving black-hole horizon behavior directly from substrate dynamics rather than introducing it phenomenologically;
- defining v_{eff} independently of observable-time circularity;
- extending latent-memory calibration beyond partial vectorial proxies;
- formalizing the arbiter as a non-anthropomorphic global coherence constraint;
- determining whether $\Lambda_{\text{eff}}(t)$ can reproduce known observational cosmology;
- deriving quantitative falsifiable astrophysical predictions;
- determining whether the proposed substrate hierarchy admits a consistent quantum-compatible formulation.

Not claimed. None of these problems are solved in the present document. The purpose of these notes is instead to organize a coherent research direction connecting substrate organization, latent reconstruction, emergent geometry, and observable cosmological dynamics.

12 Conclusion

These notes propose a cautious cosmological extension of the OMM–SBT–LIM framework.

The central idea is that the observable universe may correspond not to a fully closed fundamental system, but rather to an emergent, finite, and stabilized dynamical regime coupled to deeper latent structural organization.

Within this perspective, several phenomena become potentially compatible with a shared substrate dynamics:

- persistent non-terminal observable organization;
- emergent or scale-dependent cosmological terms;
- black holes interpreted as inter-regime attractors;
- effective gravitational amplification through structural density;
- observable geometry emerging from substrate organization.

The computational probes presented here do not validate a cosmological theory. They instead demonstrate that the proposed OMM–SBT–LIM architecture can reproduce internally coherent persistence and structural amplification behaviors within simplified numerical regimes.

Not claimed. The present framework does not establish a completed theory of cosmology, gravitation, or quantum geometry. The proposed extensions remain exploratory hypotheses

requiring substantial mathematical development, independent computational verification, relativistic consistency analysis, and empirical confrontation.

Nevertheless, these results suggest that OMM–SBT–LIM may provide a coherent research direction for studying the relations between latent structure, observable reconstruction, emergent geometry, and large-scale cosmological dynamics.

Code Availability

The computational probes, numerical experiments, and reproducibility artifacts associated with these extended notes are available at:

<https://github.com/NicolasGilbertAlbertRoux/omm-sbt-lim-cosmology.git>

The repository is released under the MIT License.

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